Pfam database: Expanding Protein Annotation Through AI-assisted Curation

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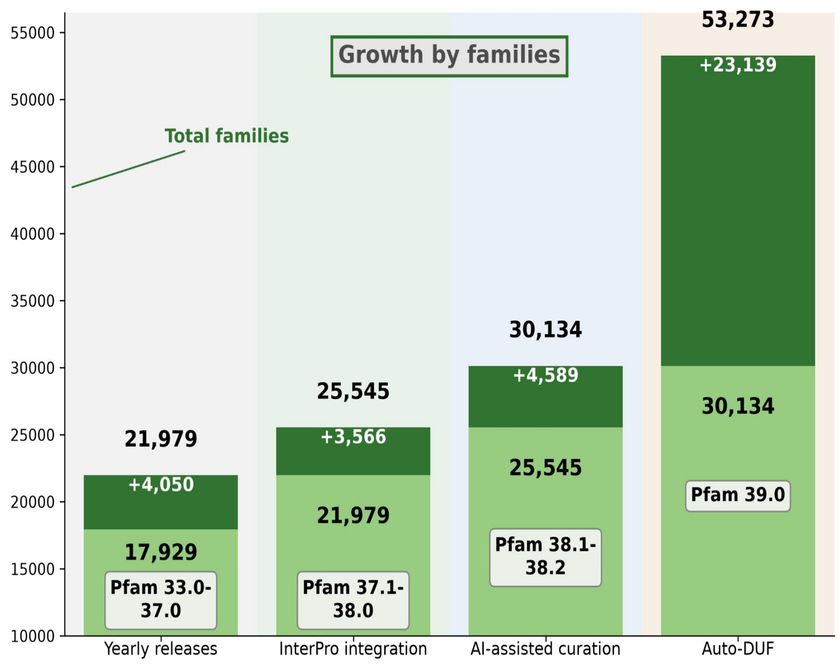
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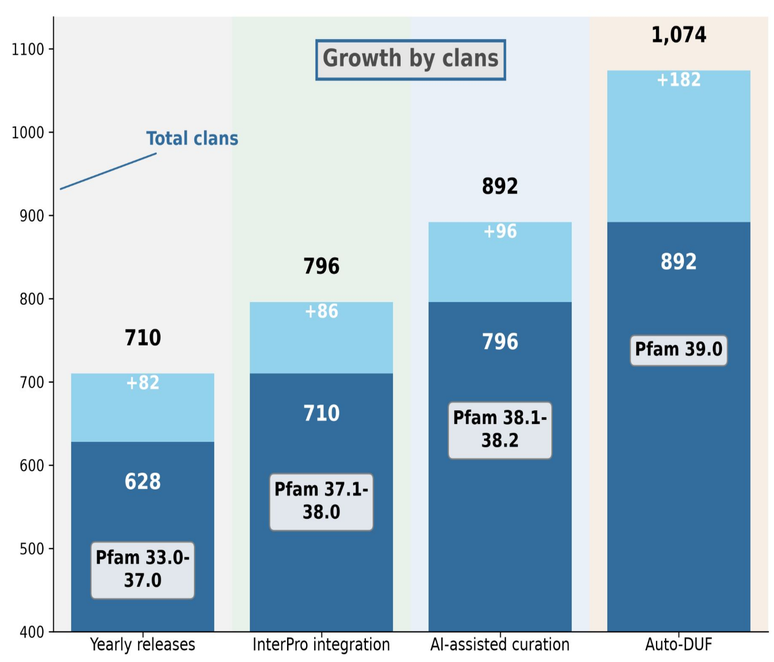
AI-assisted Curation and the Pfam expansion

Pfam is a publicly available, comprehensive repository of protein families. It is represented by manually curated multiple sequence alignments and hidden Markov models and is a key resource for biological research. In recent years, Pfam has been fully integrated into InterPro. This enabled streamlined management and more frequent updates. This transition resulted in an increase of 809 families, from 24,736 to 25,545. More recently, the introduction of AI-assisted curation has further accelerated Pfam expansion. **Here, we describe the AI-assisted curation workflow developed to support this expansion and the data to show the Pfam increase.**

Which have been the two major changes in Pfam?

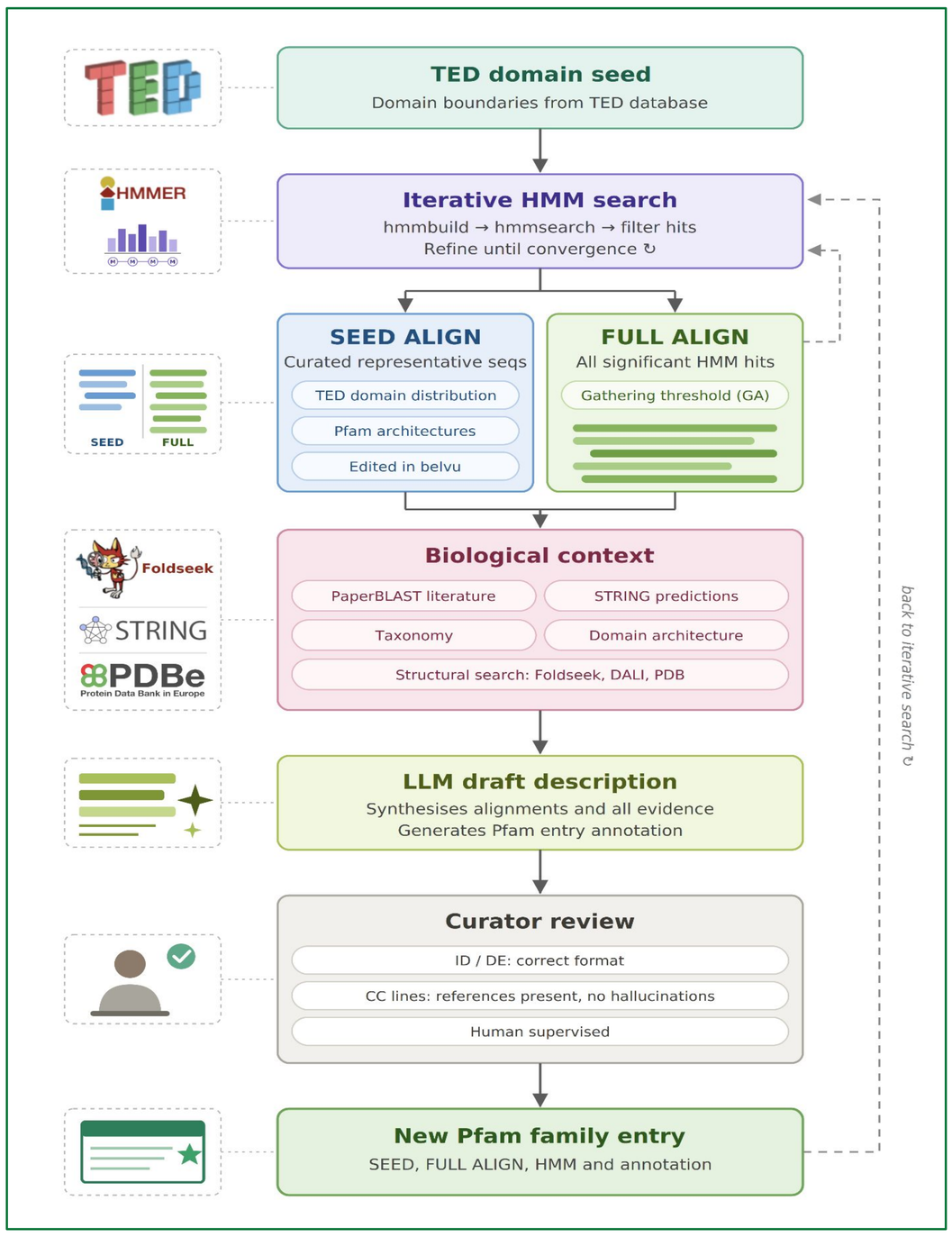
Until release 37.0, Pfam was updated yearly, adding ~800 families per release. Pfam's growth has since been driven by two major changes. First, **integration into InterPro** brought two-month releases from Pfam 37.1 onwards. Together with AlphaFold2 models and the collaboration with ECOD, this added 3,605 families between Pfam 37.1 and 38.0, almost as many as the previous five yearly releases combined. Secondly, **AI-assisted curation** added 1,936 families in Pfam 38.1 and 2,653 in 38.2, mainly by bringing all the evidence a curator needs into a single tool and by using LLM-drafted annotations to accelerate the process. Automatic annotation of uncharacterised families (auto-DUFs) further expanded this growth, adding 23,139 families in Pfam 39.0 alone and reaching 53,273 families and 1,050 clans.





The workflow: combing annotation, structural data and visibility

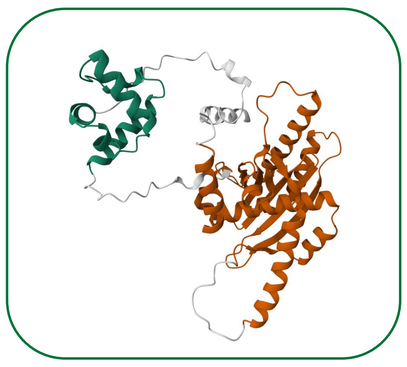
Candidate domains from TED are refined into Pfam families through iterative HMM searches. An interactive pipeline then brings all the evidence a curator needs into a single view (including SCOOP, HHsearch, Dali, Foldseek, Paperblast, String, disulphide oligomeric), and an LLM drafts the annotation for curator review.



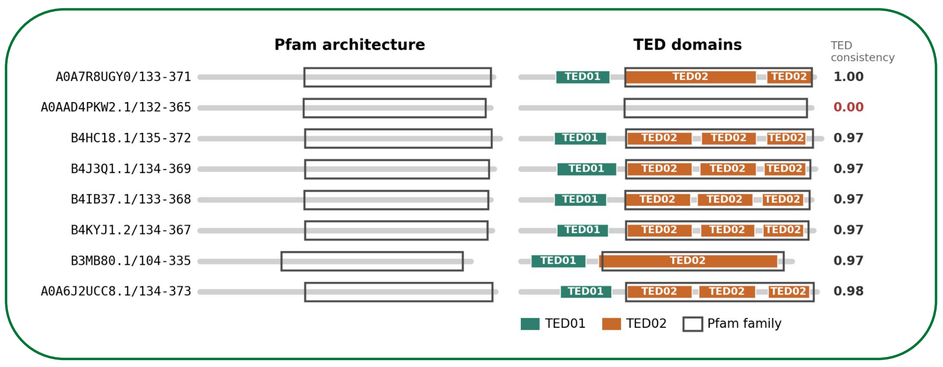
A live example and the automatic pipeline

PF43822 (Dmel-CG14450-like C-terminal domain) was built from TED domain A0A7R8UGY0_TED02. Its boundaries match TED02 across the SEED (mean TED consistency 0.85), and its hits are mostly from insects. DALI hits supported by PDB matches, placing the family in clan CL0219 (RNase H-like). The LLM drafted the description, which the curator reviewed before creating the entry.

A0A7R8UGY0_TED02

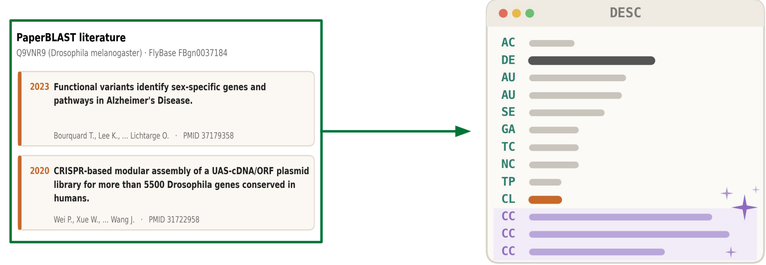


TED domain distribution and Pfam architecture



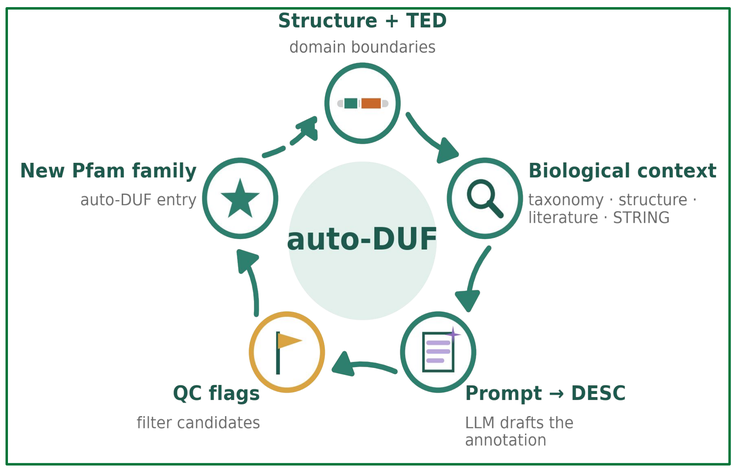
Biological context

Curator review and New Pfam entry



Auto-DUF pipeline

Once robust, the same workflow was automated for domains of unknown function. Structure, TED boundaries and biological context feed the prompt, the LLM drafts the DESC, and QC flags filter candidates before new families are created. Auto-DUFs added 23,139 families in Pfam 39.0.



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